

0	6001257	5.010625	Apartment	Entire	{ "Wireless Internet", "Air	3	1.0	Real Bed	strict	True	40.606524	73.001617	Beautiful	Brooklyne Heights	3
---	---------	----------	-----------	--------	-----------------------------	---	-----	----------	--------	------	-----------	-----------	-----------	-------------------	---

```
print(df.isnull().sum())
```

Dataset Overview:
Shape: (74111, 29)

First 5 rows:

	id	log_price	property_type	room_type	amenities	accommodates	bathrooms	bed_type	cancellation_policy	cleaning_fee	...	latitude	longitude	name	neighbourhood	number_of_reviews	review
0	6901257	5.010635	Apartment	Entire home/apt	("Wireless Internet", "Air conditioning", "Kitch...	3	1.0	Real Bed	strict	True	...	40.696524	-73.991617	Beautiful brownstone 1-bedroom	Brooklyn Heights		2
1	6304928	5.129899	Apartment	Entire home/apt	("Wireless Internet", "Air conditioning", "Kitch...	7	1.0	Real Bed	strict	True	...	40.766115	-73.989040	Superb 3BR Apt Located Near Times Square	Hell's Kitchen		6

```

# A tibble: 5 x 29
  home/apt      condition...  ...midtown DC  Heights
  <chr>         <chr>          <chr>
1 ...
2 ...
3 ...
4 ...
5 ...

5 rows x 29 columns

Data Types:
id          int64
log_price   float64

```

```

bathrooms      200
bed_type        0
cancellation_policy  0
cleaning_fee    0
city            0
description     0
first_review    15864
host_has_profile_pic    188
host_identity_verified  188
host_response_rate  18299
host_since      188
instant_bookable  0
last_review     15827
latitude        0
longitude       0
name            0
neighbourhood    6872
number_of_reviews  0
review_scores_rating  16722
thumbnail_url    8216
zipcode         968
bedrooms        91
beds            131
dtype: int64

```

```
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())
```

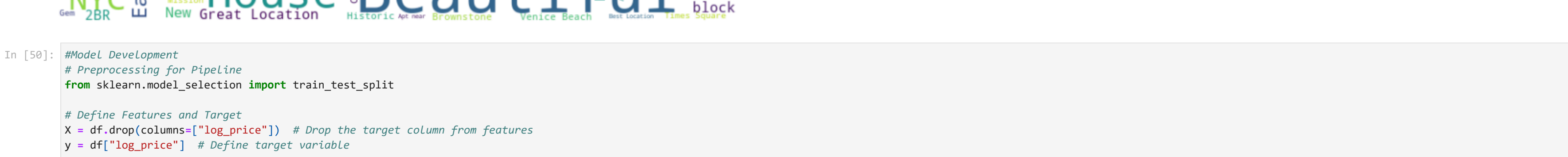
```
Missing Values Before Cleaning:
id                0
log_price         0
property_type     0
room_type         0
amenities         0
accommodates      0
```

host_identity_verified	188
host_response_rate	18299
host_since	188
instant_bookable	0
last_review	0
latitude	0
longitude	0
name	0
neighbourhood	6872
number_of_reviews	0
review_scores_rating	16722
thumbnail_url	8216
zipcode	968
bedrooms	31
beds	131

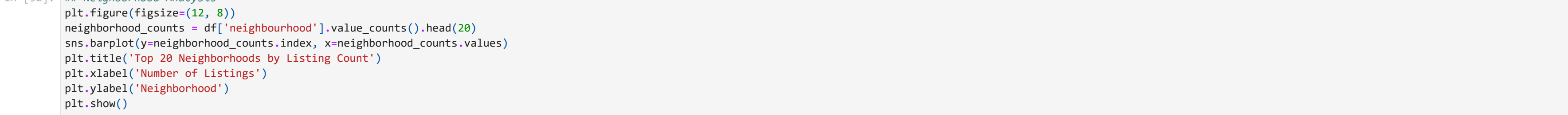
```
plt.axis('off')
plt.title('Common Words in Listing Names')
plt.show()
```

Common Words in Listing Names

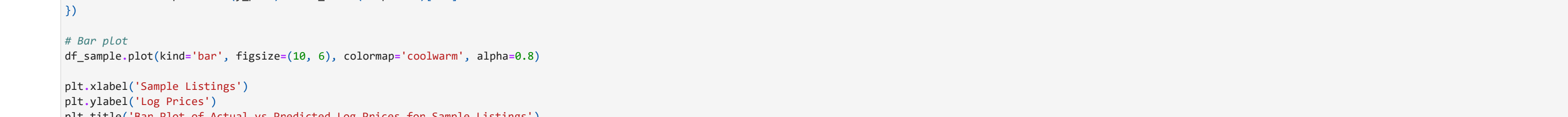
View Central Park Suite Oasis Sunny Room Flat Room Near Charming Modern Big Stylish Garden Manhattan Mediterranean Miles



```
graph TD; subgraph num; direction TB; num_SI[SimpleInputer]; num_SS[StandardScaler]; end; subgraph cat; direction TB; cat_SI[SimpleInputer]; cat_OHE[OneHotEncoder]; end; num --> XGBRegressor[XGBRegressor]; cat --> XGBRegressor;
```



Area	Very satisfied (%)
Upper West Side	95
Mid-Wilshire	90
Harlem	85
Hollywood	80



Category	Blue Bar (Log Price)	Red Bar (Log Price)
1	3.8	3.9
2	4.0	4.1
3	4.1	4.2
4	4.2	4.3
5	4.3	4.4
6	4.4	4.5
7	4.5	4.6
8	4.6	4.7
9	4.7	4.8
10	4.8	4.9
11	4.9	5.0
12	5.0	5.1
13	5.1	5.2
14	5.2	5.3
15	5.3	5.4

